Dynamic Guideline: WHO-Based Onchocerciasis - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

Scope note on WHO documents reviewed

The verifiable, disease-specific WHO guideline for onchocerciasis elimination is the **2016 WHO guideline** *Guidelines for stopping mass drug administration and verifying elimination of human onchocerciasis: criteria and procedures*. The supplied evidence did **not** verify five distinct formal WHO onchocerciasis guidelines with full GRADE methods; therefore, this report synthesizes **exactly five WHO/WHO-linked guidance documents or technical resources** relevant to onchocerciasis diagnosis, treatment, management, MDA, mapping, surveillance, and elimination, while clearly distinguishing the formal guideline from other WHO technical resources.

#	WHO guidance document/resource	Publication/ update status in retrieved evidence	Role in this report	Source
1	**Guidelines for stopping mass drug administration and verifying elimination of human onchocerciasis: criteria and procedures**	Published **1 January 2016**; NCBI record lists guideline update date **2020**	Principal formal WHO guideline for stopping MDA, post-treatment surveillance, and verification	WHO publication page, WHO IRIS PDF, NCBI

#	WHO guidance document/resource	Publication/ update status in retrieved evidence	Role in this report	Source
2	**WHO onchocerciasis fact sheet**	Current WHO technical summary available before **29 April 2026**	Current treatment, MDA, mapping, elimination milestones, country status	WHO fact sheet
3	**Global Onchocerciasis Network for Elimination**	WHO initiative page current before **29 April 2026**	WHO 2030 targets, cross-border elimination framing, guidance in process	WHO GONE
4	**Elimination mapping of onchocerciasis: WHO handbook**	WHO handbook available before **29 April 2026**	Mapping, foci definition, documentation for elimination dossiers	WHO elimination mapping handbook
5	**WHO AFRO/ESPEN NTD Master Plan guidance 2026–2030**	Planning guidance for **2026–2030**	Programme management, safety planning, serious adverse event response, loiasis co-endemic challenges	WHO AFRO/ ESPEN NTD Master Plan 2026–2030 PDF

Most recent WHO "Meaningful youth engagement" item identified

The WHO item matching the requested title pattern is **not an onchocerciasis guideline**. The verified title is **“Meaningful youth engagement in a WHO guideline development process: case study of WHO abortion care guideline (2022)”**, launched by WHO on **11 December 2025**. It is methodologically relevant to inclusive guideline development but does not provide onchocerciasis-specific recommendations WHO webinar page, WHO publication page.

Principal formal WHO onchocerciasis guideline

- **Guideline title:** *Guidelines for stopping mass drug administration and verifying elimination of human onchocerciasis: criteria and procedures*.
- **Publication date: 1 January 2016**.
- **Target population/audience:** Policy-makers in endemic countries, national neglected tropical disease programmes, national onchocerciasis elimination programmes, national oversight committees, and verification teams WHO publication page, PAHO/WHO.
- **Clinical/programmatic target population:** Populations in endemic foci receiving ivermectin MDA, especially communities approaching interruption of *Onchocerca volvulus* transmission.
- **Scope and objectives:** To guide programmes on:
    - when to stop ivermectin MDA;
    - how to conduct **post-treatment surveillance for 3–5 years**;
    - how to confirm interruption of transmission using entomological and serological evidence;
    - how to prepare for WHO verification of elimination through an International Verification Team before official acknowledgement by the WHO Director-General WHO publication page, NCBI Recommendations.

Current WHO elimination context

WHO frames onchocerciasis under the **NTD Road map 2021–2030** as targeted for elimination through interruption of transmission. By 2030, WHO targets include stopping ivermectin MDA in at least one focus in **34 countries**, stopping MDA in more than 50% of the population in at least **16 countries**, stopping MDA in the entire endemic population in at least **12 countries**, and verifying interruption of transmission in **12 countries** WHO GONE, WHO fact sheet.

WHO reports that **Colombia, Ecuador, Mexico, Guatemala, and Niger** have been verified free of onchocerciasis; Niger was verified in **2025** and is the first African country verified free of onchocerciasis WHO fact sheet, WHO GONE.

1.2 Key Recommendations

GRADE note: Where WHO explicitly reported certainty and strength, those ratings are used. Where later WHO technical resources did not provide formal GRADE tables, certainty is rated here using the standard GRADE approach: randomized evidence starts high, observational/programmatic evidence starts low, and certainty is downgraded for indirectness, imprecision, inconsistency, or operational uncertainty.

Recommendation	GRADE Evidence Quality	Strength
Use **ivermectin population-based treatment/MDA** as the core elimination strategy in endemic areas, with at least annual treatment for 10–15 years and high therapeutic coverage; WHO's 2016 MDA framework requires at least **80% therapeutic coverage** per round.	Moderate	Strong
Establish or maintain a **national independent onchocerciasis oversight/elimination committee** to review mapping, treatment, entomology, serology, recrudescence, and cross-border risks before recommending stopping MDA.	Low	Strong
Use **O-150 PCR PoolScreen testing in black flies** to demonstrate interruption of *O. volvulus* transmission before stopping MDA.	High	Strong
Apply the entomological stopping threshold: upper 95% confidence bound **<0.1%** L3 prevalence in parous flies, or **<0.05%** in all flies assuming 50% parity.	High	Strong
Use **Ov16 serology in children under 10 years** as a supportive test, particularly when O-150 PCR results are at or near the threshold; operational threshold is **<0.1% Ov16 seropositivity**.	Low	Conditional
Do **not** use assessment of ocular infection to confirm	Low	Strong

Recommendation	GRADE Evidence Quality	Strength
interruption of transmission.		
After stopping MDA, conduct **post-treatment surveillance for 3–5 years**, with possible extension based on recrudescence, lymphatic filariasis treatment status, and cross-border transmission risk.	Low	Strong
At the end of post-treatment surveillance, use **O-150 PCR in black flies** to confirm interruption of transmission; use Ov16 serology in children when PCR results are at or near threshold.	High for O-150 PCR; Low for Ov16 adjunctive use	Strong for O-150 PCR; Conditional for Ov16
Prepare a national elimination dossier after all endemic foci complete post-treatment surveillance; WHO verification requires assessment by an **International Verification Team** before acknowledgement by the WHO Director-General.	Low	Strong
In areas co-endemic for **loiasis**, adapt MDA planning to mitigate risk of serious adverse events in people with high *Loa loa* microfilaraemia.	Low	Strong
Include serious adverse event preparedness in NTD master plans: recognition, response, investigation, reporting, and prevention.	Low	Strong
Use onchocerciasis elimination mapping to identify unmapped transmission areas and determine whether MDA is required.	Low	Strong

Sources: WHO IRIS PDF, NCBI Recommendations, NCBI Executive Summary, Mectizan-hosted WHO guideline PDF, WHO fact sheet, WHO elimination mapping handbook, WHO AFRO/ESPEN NTD Master Plan 2026–2030 PDF.

1.3 Guideline Development Methodology

Evidence review process

The 2016 WHO guideline is described as the first evidence-based guideline developed by the WHO Department of Control of Neglected Tropical Diseases according to international standards. It included systematic reviews for diagnostic tests used to measure interruption of transmission and confirm elimination, including O-150 PCR, Ov16 serology, skin snip microscopy, and ocular assessment WHO publication page, Mectizan-hosted WHO guideline PDF.

The guideline's diagnostic review found that skin snip microscopy has high specificity but poor sensitivity near elimination; one summary reports sensitivity around **20%**, limiting usefulness when infection prevalence is very low Mectizan-hosted WHO guideline PDF.

Consensus method

The WHO guideline used a guideline development group/committee process to issue recommendations for stopping MDA, conducting post-treatment surveillance, and verifying elimination. Recommendations are reported with certainty terms such as high or low certainty and strength terms such as strong or conditional NCBI Recommendations, NCBI Executive Summary.

Later WHO onchocerciasis technical advisory work used structured PICO questions to evaluate alternative treatment strategies, including moxidectin, ivermectin, and doxycycline, and recommended presenting monitoring-and-evaluation and post-treatment-surveillance algorithms to country programmes for review before publication WHO Onchocerciasis Technical Advisory Subgroup report.

Conflict of interest management

The supplied summaries do not provide a detailed conflict-of-interest table for the WHO 2016 guideline. Therefore, this report does not infer undisclosed COI procedures beyond noting that WHO guideline processes generally require declaration and management of conflicts.

Stakeholder and community engagement

WHO's 2025 methodological brief on meaningful youth engagement, although focused on the 2022 WHO Abortion Care Guideline rather than onchocerciasis, provides a transferable model for future onchocerciasis guideline development: youth representatives in evidence/recommendation groups, youth-led convening, and explicit use of lived experience and values in guideline development WHO publication page, WHO webinar page. No supplied source verifies meaningful youth engagement in an onchocerciasis WHO guideline process.

II. Evidence Summary After WHO Latest Guideline Release (Guidelines for stopping mass drug administration and verifying elimination of human onchocerciasis) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

Search strategy

Evidence was synthesized from the supplied research summaries covering WHO, CDC, peer-reviewed journals, trial registries, and technical resources from **2016 through 29 April 2026**. Key search concepts included:

- onchocerciasis AND ivermectin;
- onchocerciasis AND moxidectin;
- onchocerciasis AND doxycycline;
- onchocerciasis AND Ov16;
- onchocerciasis AND O-150 PCR;
- onchocerciasis AND xenomonitoring;
- stopping MDA AND verification;
- onchocerciasis elimination mapping;
- loiasis co-endemic safety;
- WHO guideline development and youth engagement.

Databases and sources represented

- WHO official publications, fact sheets, technical pages, and WHO AFRO/ESPEN documents.
- CDC clinical care guidance.
- Peer-reviewed journals including *The Lancet* and *PLOS Neglected Tropical Diseases*.
- Clinical trial registry/protocol materials.
- WHO-linked technical advisory and programme reports.

Inclusion criteria

Included evidence had to address at least one of:

1. treatment or MDA for onchocerciasis;
2. stopping MDA or verifying elimination;
3. diagnostic tools such as O-150 PCR, Ov16 serology, skin snips, or blackfly testing;
4. post-treatment surveillance or mapping;
5. safety in loiasis co-endemic settings;
6. post-2016 treatment alternatives such as moxidectin or doxycycline.

Exclusion criteria

Excluded evidence included:

- non-onchocerciasis ivermectin evidence, such as COVID-19 ivermectin GRADE tables;
- sources without relevance to diagnosis, treatment, MDA, surveillance, or elimination;
- youth-engagement evidence unless used only as guideline-development methodology.

2.2 Key New Studies

Study/source	Design	Key findings	GRADE Quality
Phase 3 moxidectin vs ivermectin trial in Ghana, Liberia,	Randomized, double-blind, active-controlled	Moxidectin produced lower skin microfilarial density at 12 months than ivermectin; one retrieved summary	High for microfilarial-density outcome; Moderate for

Study/source	Design	Key findings	GRADE Quality
and DRC	trial	reports adjusted geometric mean **0.6 mf/mg** with moxidectin vs **4.5 mf/mg** with ivermectin, treatment difference **86%**, **p < 0.0001**. Trial interpretation: moxidectin could reduce transmission between treatment rounds more than ivermectin.	inferred transmission/ elimination impact
ClinicalTrials.gov **moxidectin safety/protocol evidence**	Randomized, double-blind, active-controlled multisite protocol	Designed to expand safety database for moxidectin 8 mg vs ivermectin and inform WHO/ministry decisions on possible mass moxidectin administration. Protocol dosing included moxidectin 8 mg for participants ≥8 years and ivermectin approximately 150 μg/kg by height.	Low for effectiveness because protocol-level evidence; Moderate for rationale
PLOS NTD diagnostic target product profile, 2022	Expert consensus/ diagnostic TPP analysis	WHO stopping-MDA framework requires serology and entomology; 0.1% Ov16 threshold is conservative but the evidence base was described as weak; cluster averages may miss transmission hotspots.	Low
CDC clinical care guidance	Authoritative clinical	Ivermectin remains treatment of choice; kills	Moderate for clinical treatment

Study/source	Design	Key findings	GRADE Quality
	guidance	microfilariae but not adult worms; one dose reduces microfilarial load for up to a year or more; no evidence prolonged daily ivermectin adds benefit over annual treatment. Doxycycline targets Wolbachia but is contraindicated in pregnancy.	principles; Low for programme extrapolation
Cameroon field study	Observational field/ programmatic study	Used skin snip microscopy to identify positives and offered doxycycline 100 mg daily for 35 days, alongside ivermectin for eligible individuals. Supports targeted test-and-treat approaches but not replacement of MDA.	Low
Togo ivermectin/ vector-control transmission study	Longitudinal observational programme study	Repeated ivermectin MDA and vector control were associated with strong reductions in onchocerciasis transmission.	Low
WHO elimination mapping handbook	WHO technical handbook	Emphasizes need to document foci boundaries, justify non-endemic status, and ensure elimination dossiers address possible transmission outside known foci.	Low

Study/source	Design	Key findings	GRADE Quality
WHO AFRO/ ESPEN NTD Master Plan 2026–2030 guidance	WHO regional programme guidance	Highlights serious adverse event preparedness and specific risks of MDA in loiasis co-endemic areas.	Low
CenterWatch/ registry listing: moxidectin vs ivermectin MDA in Angola	Trial registry listing	Ongoing/planned trial comparing moxidectin and ivermectin MDA; outcomes include prevalence/intensity, blackfly testing, parasite genetic testing, and modelling years to elimination.	Very Low for current recommendations; potentially important future evidence

Sources: Lancet/ScienceDirect trial, ClinicalTrials.gov protocol PDF, PLOS NTD TPP PDF, CDC, PLOS Neglected Tropical Diseases, PMC Togo study, WHO elimination mapping handbook, WHO AFRO/ESPEN NTD Master Plan 2026–2030 PDF, CenterWatch trial listing.

2.3 Evidence Gaps and Future Research Needs

1. **Stopping-MDA thresholds need stronger validation.**
   The WHO 0.1% Ov16 threshold remains operationally important but is described in a 2022 diagnostic TPP paper as based on weak evidence, albeit intentionally conservative to avoid premature stopping PLOS NTD TPP PDF.
2. **Hotspot detection remains a major risk.**
   Average survey-area prevalence may hide localized transmission hotspots, which could lead to stopping MDA too early PLOS NTD TPP PDF.
3. **Moxidectin is promising but not yet a universal replacement for ivermectin MDA.**
   RCT data show superior prolonged microfilarial suppression, but programme-level evidence on safety, acceptability, cost, repeated MDA rounds, pregnancy/child eligibility, loiasis co-endemic settings, and time to elimination is still needed Lancet/ScienceDirect trial, ClinicalTrials.gov protocol PDF.

4. **Doxycycline has biological plausibility and macrofilaricidal relevance but operational limits.**
   It targets Wolbachia, which supports adult worm survival and embryogenesis, but prolonged regimens and pregnancy contraindication limit mass use CDC.
5. **Loiasis co-endemic settings require safer mapping and treatment algorithms.**
   WHO identifies loiasis co-endemicity as a barrier because treatment of individuals with high *Loa loa* microfilaraemia can cause serious adverse events WHO moxidectin EML application PDF, WHO AFRO/ESPEN NTD Master Plan 2026–2030 PDF.
6. **Community and youth engagement are underdeveloped in onchocerciasis guideline processes.**
   WHO has documented meaningful youth engagement in the abortion-care guideline process, but no supplied source verifies an analogous onchocerciasis process WHO publication page, WHO webinar page.

2.4 Updated Recommendations Based on New Evidence

Original Recommendation	New Evidence Impact	Updated Recommendation
Use ivermectin MDA as the main treatment strategy for endemic communities.	CDC and WHO continue to support ivermectin as the treatment/MDA standard; ivermectin kills microfilariae but not adult worms.	**Continue ivermectin MDA** as standard of care and public-health strategy, aiming for ≥80% therapeutic coverage and at least annual rounds until stopping criteria are met.
Use O-150 PCR in black flies to determine whether MDA can stop.	No stronger replacement test has superseded O-150 PCR; diagnostic literature confirms its central role but highlights operational	**Maintain O-150 PCR xenomonitoring** as the primary stopping-MDA and post-treatment confirmation tool.

Original Recommendation	New Evidence Impact	Updated Recommendation
	challenges.	
Use Ov16 serology in children as supportive evidence.	2022 diagnostic TPP work questions the evidence base for the 0.1% threshold but recognizes its conservative purpose.	**Retain Ov16 serology** as an adjunctive tool, especially when entomological results are near threshold, but interpret with attention to sampling design, hotspot risk, and confidence intervals.
Conduct post-treatment surveillance for 3–5 years.	No evidence supports shortening surveillance. Cross-border risk and hotspot concerns argue for caution.	**Continue 3–5 years of post-treatment surveillance**, extending where cross-border transmission, recrudescence risk, or weak surveillance exists.
Do not use ocular infection assessment to confirm interruption.	No new evidence supports reinstating ocular assessment as a stopping criterion.	**Do not use ocular infection assessment** to confirm transmission interruption.
Ivermectin is the principal MDA drug.	Moxidectin RCTs show superior sustained microfilarial suppression compared with ivermectin.	**Do not replace ivermectin routinely yet**. Consider moxidectin only where national/WHO policy permits, safety data are adequate, and implementation conditions are met; prioritize trials and operational research.
Skin snip microscopy may be used in some field contexts.	Low sensitivity near elimination limits usefulness for stopping decisions.	**Avoid relying on skin snips for stopping MDA**; reserve for clinical/ targeted contexts or research where appropriate.
Standard MDA in endemic	Loiasis co-endemicity poses risk of severe	**In loiasis co-endemic areas**, require risk stratification, serious adverse event

Original Recommendation	New Evidence Impact	Updated Recommendation
communities.	adverse events.	preparedness, and adapted treatment strategies before MDA.
Guideline development should follow WHO evidence standards.	WHO’s 2025 youth-engagement brief provides a transferable model for values-and-preferences input.	**Future onchocerciasis WHO guidelines should add structured community, youth, and affected-population engagement**, especially for adherence, acceptability, stigma, surveillance, and post-MDA implementation.

Consolidated Clinical and Programmatic Guidance

Diagnosis and monitoring

- For **elimination programmes**, prioritize:
  - **O-150 PCR PoolScreen testing in black flies** during peak transmission season;
  - **Ov16 serology in children under 10 years**, especially when entomological findings are near threshold;
  - statistically valid sample sizes, correct target populations, and confidence intervals NCBI Background, Mectizan-hosted WHO guideline PDF.
- Do not use ocular infection assessment to confirm interruption of transmission NCBI Executive Summary.
- Skin snips may have clinical or research utility but are insufficiently sensitive for near-elimination stopping decisions Mectizan-hosted WHO guideline PDF.

Treatment

- **Ivermectin** remains the standard treatment and MDA drug. It kills microfilariae, reduces blindness and skin disease, and substantially reduces microfilarial load for up to a year or more, but it does not kill adult worms CDC.

- **Moxidectin** has stronger sustained microfilarial suppression than ivermectin in RCT evidence, but broader MDA policy adoption requires safety, implementation, and programme evidence Lancet/ScienceDirect trial, ClinicalTrials.gov protocol PDF.
- **Doxycycline** targets Wolbachia, which supports adult worm survival and embryogenesis, but it is contraindicated in pregnancy and operationally less suited to mass use because of prolonged dosing CDC.

MDA stopping and verification

A programme should stop MDA only when:

1. entomological evidence demonstrates interruption using O-150 PCR thresholds;
2. Ov16 serology supports interruption where indicated;
3. data are statistically valid with appropriate confidence intervals;
4. the national oversight committee has reviewed all relevant risks;
5. cross-border and recrudescence risks have been addressed.

After stopping MDA:

- conduct **3–5 years** of post-treatment surveillance;
- confirm interruption at the end of surveillance;
- submit the country dossier after all endemic foci complete surveillance;
- proceed to WHO verification through an International Verification Team before official acknowledgement by the WHO Director-General WHO publication page, NCBI Executive Summary.

Final Dynamic Guideline Position as of 29 April 2026

The 2016 WHO guideline remains the controlling disease-specific WHO guidance for stopping MDA and verifying elimination of human onchocerciasis. Post-2016 evidence supports continuing ivermectin MDA and WHO's O-150 PCR/Ov16 surveillance framework, while highlighting three areas requiring caution: uncertainty around conservative serological thresholds, risk of missed hotspots, and safety challenges in loiasis co-endemic areas. Moxidectin is the most important emerging alternative because it produces more sustained microfilarial suppression than ivermectin, but available evidence does not yet justify universal replacement of ivermectin MDA outside policy-approved or research settings. Future WHO onchocerciasis guideline updates should incorporate stronger diagnostic validation, operational evidence for moxidectin, loiasis-safe strategies, cross-border surveillance, and structured engagement of affected communities and youth.